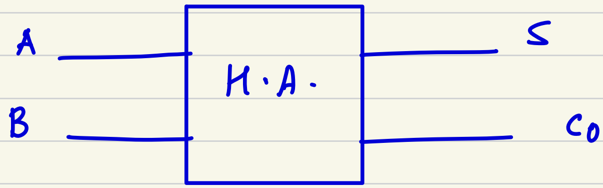
TRUTH TABLES, K-MAPS & SIMPLI-FIED BOOLEAN EXPRESSION

1. Half Adder



Truth Table

A	B	S	C_o
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

K- Map for Sum

A \ B	0	1
0	(0)	1 (1)
1	1 (2)	(3)

$$\boxed{S = A\bar{B} + \bar{A}B = A \oplus B}$$

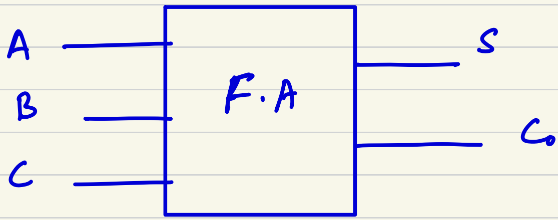
Simplified Boolean exp.

K-Map for C_o

A \ B	0	1
0	$_0$	$_1$
1	$_2$	1 $_3$

$$\boxed{C_o = AB}$$

2. Full Adder



Truth Table

A	B	C	S	C_o
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

K-Map for Sum (S)

A \ BC	00	01	11	10
0	$_{0}$	1_{1}	$_{3}$	1_{2}
1	1_{4}	$_{5}$	1_{7}	$_{6}$

$$S = A\bar{B}\bar{C} + \bar{A}\bar{B}C + ABC + \bar{A}B\bar{C}$$

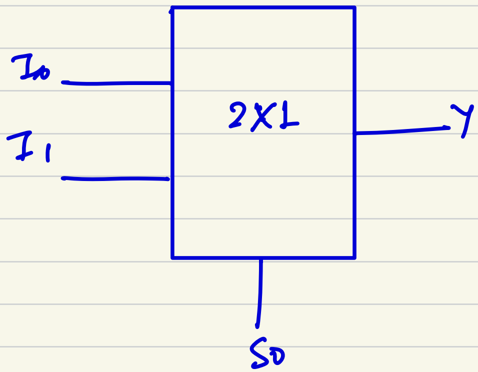
$$S = A \oplus B \oplus C$$

K-Map for C_o

A \ BC	00	01	11	10
0	$_{0}$	$_{1}$	1_{3}	$_{2}$
1	$_{4}$	1_{5}	1_{7}	1_{6}

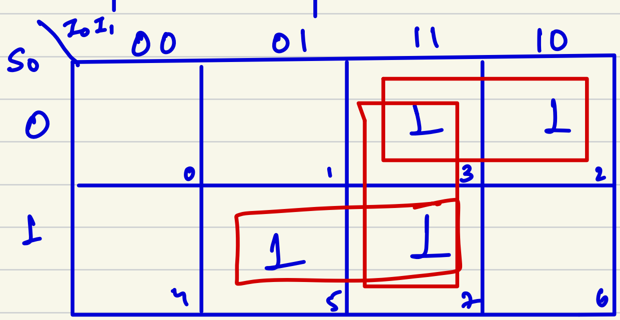
$$C_o = BC + AC + AB$$

3. 2 : 1 MUX



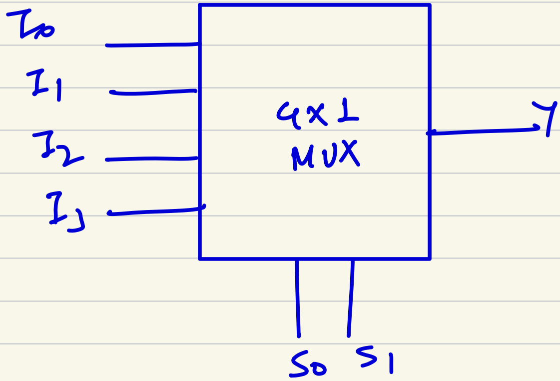
Truth Table

S_0	I_0	I_1	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1



$$Y = \overline{S_0} I_0 + S_0 I_1$$

4. 4x1 MUX



S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Boolean expression for Y

$$Y = \overline{S_1}\,\overline{S_0} I_0 + \overline{S_1} S_0 I_1 + S_1 \overline{S_0} I_2 + S_1 S_0 I_3$$